Music preference classification

A comparison of five classification models

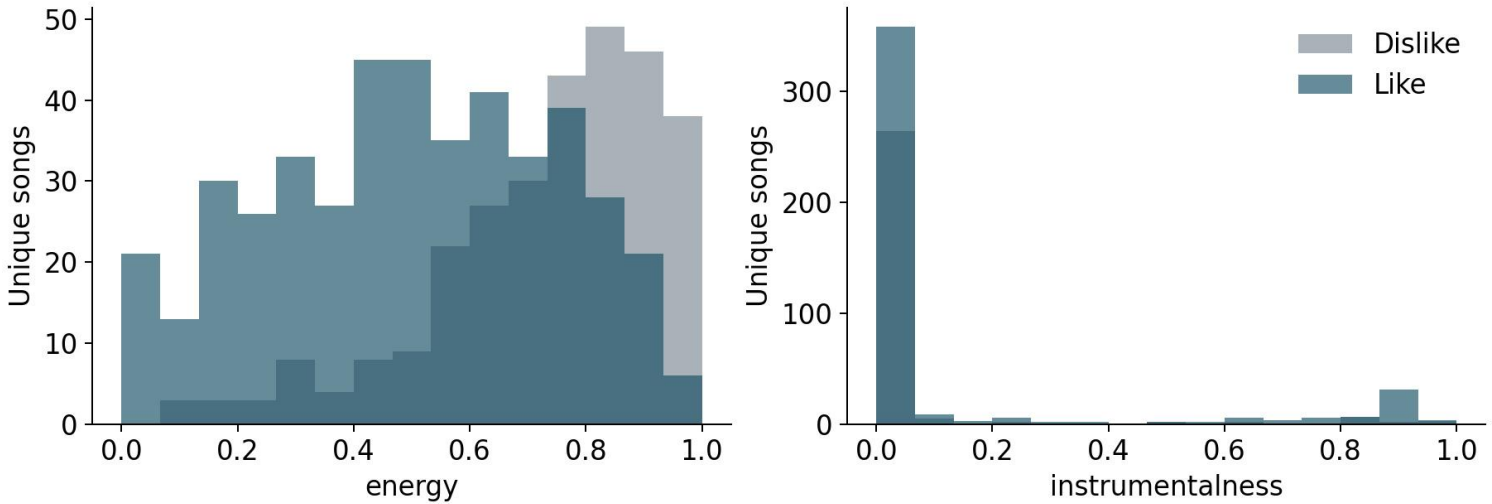
750 labeled rows 13 features 200 target songs

Lab 1
Presenter: XINRAN LI

Dataset inspection

736 records after removing 14 duplicate feature rows

443 likes / 293 dislikes No missing or nonfinite values



The two classes overlap in energy and instrumentalness.

Features and preprocessing

Inputs	Treatment and motivation
10 numeric features	StandardScaler for logistic regression, k-NN and SVM Original scale for random forest and gradient boosting
key, mode, time_signature	One-hot encoding Avoid imposing numeric distances on category codes
Feature selection / outliers	Keep all 13 inputs to limit additional model search Retain plausible extreme values

Numeric: acousticness, danceability, duration, energy, instrumentalness, liveness, loudness, speechiness, tempo and valence

Validation and model selection

Fixed split

588 development records and 148 holdout records
Stratified sampling with random seed 42

Shared five-fold cross-validation

Use the same development folds for every model and setting
Fit preprocessing within each fold’s training portion

Selection rule

Choose the highest mean CV accuracy
Freeze the choice before evaluating the holdout

Logistic regression and k-NN

Logistic regression

A weighted feature score becomes a probability through sigmoid
A linear boundary in the encoded feature space
C controls regularization strength

k-nearest neighbors

Nearby training records vote with equal or distance-based weights
Sensitive to feature scales, irrelevant inputs and neighbor count k

Both methods standardize numeric inputs and encode categories.

Support vector machine with RBF kernel

Classification boundary

Seek a large margin between classes
Support vectors determine the boundary

RBF kernel

Capture nonlinear relationships between song features
Standardize numeric inputs and encode categories

Tuning

C controls the penalty for margin violations
gamma controls how local the kernel similarity is

Random forest and gradient boosting

Random forest

Train trees on bootstrap samples with random candidate features
Combine their predicted class probabilities
Depth and minimum leaf size limit complexity

Gradient boosting

Add shallow trees sequentially to reduce prediction loss
Learning rate, tree count and depth work together

Both methods encode categories and retain numeric feature scales.

Hyperparameter search

Model	Search space	Settings
Logistic regression	C: 0.01, 0.1, 1, 10	4
k-NN	k: 3, 5, 9, 15, 25, 41 Weights: uniform / distance	12
RBF SVM	C: 0.1, 1, 10, 100 gamma: scale, 0.01, 0.1	12
Random forest	150 trees. Depth: 4, 8, unrestricted Minimum leaf size: 1, 3, 8	9
Gradient boosting	Trees: 100 / 200. Rate: 0.03 / 0.1 Depth: 1 / 2	8

45 settings × 5 folds = 225 validation fits. Tuning metric: accuracy.

Cross-validation comparison

Mean five-fold accuracy



Majority baseline: 60.20%. CV scores include tuning optimism.

Random forest leads gradient boosting by only 0.33 percentage points.

Selected model: holdout evaluation

Random forest: 120/148 correct = 81.08%

Majority baseline: 60.14%

Actual / predicted	Dislike (0)	Like (1)
Dislike (0)	41	18
Like (1)	10	79

Balanced accuracy 79.13% Like-class F1 0.849
ROC-AUC 0.900

The holdout contains only 148 songs, so accuracy may vary on new data.

Final training and submission

Selected settings

Random forest with 150 trees
max_depth = 8 and min_samples_leaf = 3

Final training

Keep the selected settings and fit all 736 deduplicated records
Predict 200 target songs in their original CSV order

Output

128 likes and 72 dislikes
results/submission.txt contains one string of 200 binary digits

Conclusions, limitations and AI use

Conclusion

Random forest achieved the highest mean CV accuracy: 83.33%
Holdout accuracy: 81.08%

Limitations

Small dataset, one holdout split and small gaps between models
Artist IDs are unavailable. Target accuracy remains unknown

AI assistance

I led the experiment design, model selection and result analysis
ChatGPT/Codex assisted with the code framework, checks and report drafting